

- * TestimonialCarousel – Rotating user testimonials
- **Signup Page** (app/signup/page.tsx)
 - * SignupForm – Registration with role selection
- **Dashboard Page** (app/dashboard/page.tsx)
 - * FarmerDashboard – Farmers-only content
 - StatCard – Statistics display cards
 - PhotoUploader – Image upload component
 - LocationPicker – GPS coordinate capture
 - PaymentButton – Razorpay integration
 - ListingCard – Individual listing display
 - * AgentDashboard – Agents-only content
 - StatCard – Statistics display cards
 - TaskCard – Available/claimed task cards
- **Admin Page** (app/admin/page.tsx)
 - * AdminStatCards – Platform-wide statistics
 - * UserManagement – User list with role editing
 - * ListingManagement – All listings with recycling controls
 - * AnalyticsCharts – Recharts-based data visualizations
- **Shared Components** (components/ui/)
 - * Button, Dialog, Card, Badge – Radix UI primitives
 - * Select, Input, Label – Form elements
 - * Toast, Tabs, Sheet – Interactive elements

7.9 Security Architecture

The security architecture of AgriTrace operates at multiple levels:

1. **Transport Layer:** All communication between the client and Firebase services is encrypted using TLS 1.3. The application is served over HTTPS, ensuring data confidentiality in transit.
2. **Authentication Layer:** Firebase Authentication manages user identity with industry-standard practices including secure password hashing (bcrypt), session token management, and automatic token refresh. Password reset flows use time-limited, single-use tokens sent via email.
3. **Authorization Layer:** Firestore Security Rules enforce role-based access control (RBAC) at the database level. Rules check the authenticated user's UID and role before allowing read/write operations. Key rules include: